

Advanced Dynamics — Equation Sheet

Kinematics · Newton's Laws · Energy & Momentum · Generalized Coordinates & Constraints · D'Alembert's Principle · Lagrangian & Hamiltonian Mechanics · Nonlinear Dynamics & Bifurcations · 2-D Linear Systems · Nondimensionalization

Notation

- $N, A, B, \dots$ — reference frames (e.g. N = fixed/inertial)
- $\hat{n}_i, \hat{a}_i, \hat{b}_i$ — right-handed unit-vector basis fixed in N, A, B
- $\vec{r}_{P/O}$ — position of point P relative to point O ; $\vec{r} \equiv {}^N\vec{r}$ unless noted otherwise
- ${}^N\vec{v}_P, {}^N\vec{a}_P$ — velocity/acceleration of P as seen from N
- $\vec{\omega}^{A/N}$ — angular velocity of frame A relative to N
- $\vec{\alpha}^{A/N} = {}^N\frac{d}{dt}\vec{\omega}^{A/N}$ — angular acceleration of A relative to N
- ${}^N\frac{d}{dt}(\cdot)$ — time derivative taken by an observer fixed in N
- $[R]^{B/A}$ — rotation (direction cosine) matrix: $\{\hat{b}\} = [R]^{B/A}\{\hat{a}\}$
- $q_1, \dots, q_N$ — generalized coordinates; $\tilde{\gamma}_{ij} = \partial\vec{r}_i/\partial q_j$ — partial velocities
- $\vec{P} = m\vec{v}$ — linear momentum; $\vec{H}_O = \vec{r} \times \vec{P}$ — angular momentum about O
- $L = T - V$ — Lagrangian; $p_j = \partial L/\partial \dot{q}_j$ — conjugate (generalized) momentum
- $H = \sum_j p_j \dot{q}_j - L$ — Hamiltonian; λ_k — Lagrange multiplier for constraint ϕ_k

Rotations & Angular Velocity

Rotation matrices

Orthogonal, so $[R]^{-1} = [R]^T$; composing rotations through an intermediate frame A :

$$[R]^{B/N} = [R]^{B/A}[R]^{A/N}$$

Angular velocity addition (chain rule)

Valid for any chain of intermediate frames:

$$\vec{\omega}^{B/N} = \vec{\omega}^{B/A} + \vec{\omega}^{A/N}, \quad \vec{\omega}^{C/N} = \vec{\omega}^{C/B} + \vec{\omega}^{B/A} + \vec{\omega}^{A/N}$$

Note the reversal/cancellation identities:

$$\vec{\omega}^{A/N} = -\vec{\omega}^{N/A}, \quad \vec{\omega}^{A/A} = \vec{0}$$

Derivative of a rotating unit vector

If $\hat{b}_i$ is fixed in B (constant length/orientation relative to B):

$${}^N\frac{d}{dt}\hat{b}_i = \vec{\omega}^{B/N} \times \hat{b}_i$$

2D rotation example (N, P frames offset by α)

$$\hat{n}_1 = \hat{p}_1 \cos \alpha + \hat{p}_2 \sin \alpha, \quad \hat{n}_2 = -\hat{p}_1 \sin \alpha + \hat{p}_2 \cos \alpha$$

$$\begin{Bmatrix} \hat{n}_1 \\ \hat{n}_2 \end{Bmatrix} = [R]^{N/P} \begin{Bmatrix} \hat{p}_1 \\ \hat{p}_2 \end{Bmatrix}, \quad [R]^{N/P} = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$$

(written ${}^N R_P$ in lecture)

$$[R]^{P/N} = ([R]^{N/P})^{-1} = ([R]^{N/P})^T = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$$

Vector Differentiation Formula (VDF)

Also called the *transport theorem*: for *any* vector $\vec{q}$ (not necessarily fixed in either frame), relating the rate of change seen in N to the rate of change seen in a frame A rotating at $\vec{\omega}^{A/N}$:

$${}^N\frac{d}{dt}\vec{q} = {}^A\frac{d}{dt}\vec{q} + \vec{\omega}^{A/N} \times \vec{q}$$

Corollary: if $\vec{q}$ is fixed in A (i.e. ${}^A\frac{d}{dt}\vec{q} = 0$), this reduces to ${}^N\frac{d}{dt}\vec{q} = \vec{\omega}^{A/N} \times \vec{q}$ (previous box). This is the tool used to differentiate a vector expressed in a rotating basis one component at a time.

Position Vector Addition (Setup for Moving Frames)

For two points P and O' both located relative to a common origin O (equivalently $\vec{r} = \vec{R} + \vec{r}'$, with $\vec{R} = \vec{r}_{O'/O}$, $\vec{r}' = \vec{r}_{P/O'}$), position

vectors add/subtract exactly like any vectors — the basic relation used to move between frames whose origins differ:

$$\vec{r}_{P/O} = \vec{r}_{P/O'} + \vec{r}_{O'/O} \quad \Longleftrightarrow \quad \vec{r}_{P/O'} = \vec{r}_{P/O} - \vec{r}_{O'/O}$$

Chained over several intermediate points: $\vec{r}_{P/O} = \vec{r}_{P/B} + \vec{r}_{B/A} + \vec{r}_{A/O}$. Applying the VDF to this relation term-by-term is exactly how the velocity/acceleration relations below are derived.

Velocity: Translating & Rotating Frames

Let O' be the (moving) origin of frame A , which translates *and* rotates ($\vec{\omega}^{A/N}$) relative to N , and let P be an arbitrary point:

$${}^N\vec{v}_P = {}^N\vec{v}_{O'} + {}^A\vec{v}_P + \vec{\omega}^{A/N} \times \vec{r}_{P/O'}$$

- ${}^N\vec{v}_{O'}$ — translational velocity of the moving origin
- ${}^A\vec{v}_P = {}^A\frac{d}{dt}\vec{r}_{P/O'}$ — velocity of P relative to A (as seen by an observer riding on A)
- $\vec{\omega}^{A/N} \times \vec{r}_{P/O'}$ — velocity contribution from A 's rotation

Special cases

- Pure translation ($\vec{\omega}^{A/N} = 0$): ${}^N\vec{v}_P = {}^N\vec{v}_{O'} + {}^A\vec{v}_P$
- P fixed in A (${}^A\vec{v}_P = 0$, rigid-body point): ${}^N\vec{v}_P = {}^N\vec{v}_{O'} + \vec{\omega}^{A/N} \times \vec{r}_{P/O'}$
- Common origin ($O' = O$, ${}^N\vec{v}_{O'} = 0$): reduces to the transport theorem applied to $\vec{r}_{P/O}$

Acceleration: Translating & Rotating Frames

Differentiating the velocity relation again (the “five-term” / Coriolis theorem):

$${}^N\vec{a}_P = {}^N\vec{a}_{O'} + {}^A\vec{a}_P + \vec{\alpha}^{A/N} \times \vec{r}_{P/O'} + \vec{\omega}^{A/N} \times (\vec{\omega}^{A/N} \times \vec{r}_{P/O'}) + 2\vec{\omega}^{A/N} \times {}^A\vec{v}_P$$

- ${}^N\vec{a}_{O'}$ — translational acceleration of the moving origin
- ${}^A\vec{a}_P = {}^A\frac{d}{dt}{}^A\vec{v}_P$ — acceleration of P relative to A
- $\vec{\alpha}^{A/N} \times \vec{r}_{P/O'}$ — Euler (angular acceleration) term
- $\vec{\omega}^{A/N} \times (\vec{\omega}^{A/N} \times \vec{r}_{P/O'})$ — centripetal term
- $2\vec{\omega}^{A/N} \times {}^A\vec{v}_P$ — Coriolis term

Special cases

- Pure translation ($\vec{\omega}^{A/N} = 0$): ${}^N\vec{a}_P = {}^N\vec{a}_{O'} + {}^A\vec{a}_P$
- P fixed in A (${}^A\vec{v}_P = {}^A\vec{a}_P = 0$, rigid-body point):

$${}^N\vec{a}_P = {}^N\vec{a}_{O'} + \vec{\alpha}^{A/N} \times \vec{r}_{P/O'} + \vec{\omega}^{A/N} \times (\vec{\omega}^{A/N} \times \vec{r}_{P/O'})$$

- P moving in A but O' fixed in N (common origin, no translation): drop the ${}^N\vec{a}_{O'}$ term — this is the form used for rotating-basis problems (e.g. spherical pendulum, rotating tube)

Newton's Laws of Motion

$$\text{1st: } \sum \vec{F} = 0 \Rightarrow {}^N\ddot{\vec{r}} = \text{const} \quad \text{2nd: } \sum \vec{F} = m {}^N\ddot{\vec{r}}$$

$$\text{3rd: } \vec{F}_{12} = -\vec{F}_{21} \quad (\text{action–reaction pairs})$$

Newton's 2nd law **requires** the acceleration relative to an inertial (Newtonian) frame N — never a rotating/accelerating frame directly.

Work & Energy

Kinetic & potential energy

$$T = \frac{1}{2}m|\dot{\vec{r}}|^2 = \frac{1}{2}m\dot{\vec{r}} \cdot \dot{\vec{r}}, \quad V = V(\vec{r})$$

$$\vec{F} = -\nabla V(\vec{r}) \quad (\text{conservative/“potential” force})$$

Common potentials

- Uniform gravity: $V = mgh$, $\vec{F} = -mg\hat{n}_3$
- Newtonian gravity: $V = -\frac{GmM}{r}$, $\vec{F} = \frac{GmM}{r^2}\hat{b}_1$ ($\hat{b}_1$: unit vector toward the attracting mass)
- Linear spring (stretch x): $V = \frac{1}{2}kx^2$, $\vec{F} = -kx\hat{n}_1$

Work–energy theorem

$$W = \int_{\vec{r}_1}^{\vec{r}_2} \vec{F} \cdot d\vec{r}, \quad T_2 - T_1 = W$$

All forces conservative ($\vec{F} = -\nabla V$):

$$T_1 + V_1 = T_2 + V_2 \quad (\text{energy conserved})$$

Only holds when *every* force doing work is conservative — friction or other non-conservative applied forces break it.

Momentum**Linear momentum**

$$\vec{P} = m\dot{\vec{r}}, \quad \dot{\vec{P}} = m\ddot{\vec{r}} = \sum \vec{F}$$

If $\sum \vec{F} = \vec{0}$, $\vec{P}$ is conserved.

Angular momentum (about inertial fixed point O)

$$\vec{H}_O = \vec{r} \times \vec{P}, \quad \dot{\vec{H}}_O = \vec{r} \times \sum \vec{F} = \sum \vec{M}_O$$

“Rotational Newton’s 2nd law.” If $\sum \vec{M}_O = \vec{0}$, $\vec{H}_O$ is conserved.

Generalized Coordinates

- **Generalized coordinates** $q_1, \dots, q_N$: minimal (or redundant) geometric variables (angles/distances) describing the configuration of a system
- **Configuration space**: the set of all configurations the system can take on, parametrized by the q_j
- A system may be described by a **redundant set** (more coordinates than DOF, related by constraint equations) or a **minimal set** (one coordinate per DOF, no constraints left)

$$n_{\text{DOF}} = \#(\text{gen. coords}) - \#(\text{indep. constraints})$$

Always reduce to the minimal set before deriving equations of motion.

Holonomic Constraints

Position-based: with N generalized coordinates and M constraints ($j = 1, \dots, M$):

$$\phi_j(q_1, \dots, q_N) = \phi_j(\vec{q}) = 0, \quad \text{or, if time-dependent, } \phi_j(\vec{q}, t) = 0$$

- **Scleronomic**: no explicit t — time-independent constraint
- **Rheonomic**: explicit t dependence — e.g. a prescribed moving support
- Each independent holonomic constraint removes exactly one DOF (box above)
- **Always rewritable as non-holonomic**: since $\phi_j \equiv 0 \Rightarrow \dot{\phi}_j \equiv 0$,

$$\dot{\phi}_j(\vec{q}, t) = \sum_{i=1}^N \frac{\partial \phi_j}{\partial q_i} \dot{q}_i + \frac{\partial \phi_j}{\partial t} = 0 \quad (a_{ji} \equiv \partial \phi_j / \partial q_i)$$

Non-Holonomic Constraints

Velocity-based: $\phi_j(\vec{q}, \dot{\vec{q}}, t) = 0$, usually taken *linear* in the velocities (Pfaffian form):

$$\phi_j = \sum_{i=1}^N a_{ji}(\vec{q}, t) \dot{q}_i + a_{jt}(\vec{q}, t) = 0$$

(no explicit t : $\phi_j = \sum_i a_{ji}(\vec{q}) \dot{q}_i = 0$)

- Every holonomic constraint can be written this way (box above); the **converse is false** — a Pfaffian constraint is genuinely non-holonomic unless it is exact/integrable (or made exact with an integrating factor; check via the Frobenius condition)
- Non-holonomic constraints reduce the number of independent *velocities* but **not** the dimension of the configuration space
- **Example — knife-edge/skate/wheel constraint**: contact-point velocity must be perpendicular to the blade, $\vec{v}_A \cdot \hat{n} = 0$ with $\hat{n} = (-\sin \theta, \cos \theta)$:

$$\phi = -\sin \theta \dot{x} + \cos \theta \dot{y} = 0$$

With $(q_1, q_2, q_3) = (x, y, \theta)$: $a_1 = -\sin q_3$, $a_2 = \cos q_3$, $a_3 = 0$. Cannot be integrated to $\phi(\vec{q}) = 0$ — genuinely non-holonomic.

D’Alembert’s Principle**Partial velocities**

For particle i with position $\vec{r}_i(\vec{q}, t)$, the partial velocity w.r.t. generalized speed $\dot{q}_j$:

$$\vec{\gamma}_{ij} = \frac{\partial \vec{r}_i}{\partial q_j} = \frac{\partial \dot{\vec{r}}_i}{\partial \dot{q}_j}$$

(equal because $\dot{\vec{r}}_i$ is linear in the $\dot{q}_j$ ’s — fastest computed by differentiating the already-known $\vec{r}_i$ directly w.r.t. $\dot{q}_j$.)

Equations of motion

For a minimal set $q_1, \dots, q_M$ and particles $i = 1, \dots, n$:

$$\sum_{i=1}^n (\vec{F}_{Ai} - m_i \ddot{\vec{r}}_i) \cdot \vec{\gamma}_{ij} = 0, \quad j = 1, \dots, M$$

- $\vec{F}_{Ai}$ — *applied* (impressed) forces on particle i only (gravity, springs, motors, ...)
- **Ideal constraint forces are excluded** (rod tension, normal forces, frictionless-contact forces) — they do no virtual work consistent with the constraints and drop out automatically
- One scalar equation per generalized coordinate — exactly M equations for M DOF
- Equivalently, with generalized force $Q_j = \sum_i \vec{F}_{Ai} \cdot \vec{\gamma}_{ij}$: $\sum_i m_i \ddot{\vec{r}}_i \cdot \vec{\gamma}_{ij} = Q_j$

Recipe

1. Choose a minimal set $q_1, \dots, q_M$ consistent with all holonomic constraints
2. Write $\vec{r}_i(\vec{q}, t)$ for every particle; differentiate for $\dot{\vec{r}}_i, \ddot{\vec{r}}_i$
3. Compute $\vec{\gamma}_{ij} = \partial \vec{r}_i / \partial q_j$ for every particle/coordinate pair
4. Identify applied forces $\vec{F}_{Ai}$ on each particle (*exclude* constraint forces)
5. For each j : $\sum_i (\vec{F}_{Ai} - m_i \ddot{\vec{r}}_i) \cdot \vec{\gamma}_{ij} = 0 \Rightarrow M$ equations of motion
6. For linear (spring/mass) systems, these assemble into $[M]\ddot{\vec{q}} + [K]\vec{q} = \vec{F}$

Lagrangian Mechanics**Euler–Lagrange equations**

For a minimal set $q_1, \dots, q_M$; equivalent to D’Alembert’s principle above (substitute $\vec{\gamma}_{ij} = \partial \vec{r}_i / \partial q_j$ into $L = T - V$ and simplify):

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) - \frac{\partial L}{\partial q_j} = Q_j^{nc}, \quad j = 1, \dots, M$$

- Q_j^{nc} — generalized force from any applied force *not* already folded into V (a prescribed drive $F(t)$, friction, ...); $= 0$ if every applied force is conservative
- One 2nd-order ODE per DOF; ideal constraint forces are still automatically excluded, since $L(\vec{q}, \dot{\vec{q}}, t)$ is built from the minimal coordinates
- Cross terms quadratic in velocity (Coriolis-type, from $\partial T / \partial q_j$) fall out of $\frac{d}{dt}(\partial L / \partial \dot{q}_j) - \partial L / \partial q_j$ automatically — no need to track them separately as in Newtonian/D’Alembert bookkeeping

Matrix form & small oscillations

Multi-DOF equations of motion assemble into

$$M(\vec{q})\ddot{\vec{q}} + \vec{C}(\vec{q}, \dot{\vec{q}}) + \vec{K}(\vec{q}, t) = \vec{0}$$

(M : symmetric positive-definite mass/inertia matrix; $\vec{C}$: Coriolis/centripetal-type $\dot{q}^2$ terms; $\vec{K}$: stiffness/gravity/forcing terms). Linearizing about an equilibrium ($\sin(\cdot) \rightarrow (\cdot)$, $\cos(\cdot) \rightarrow 1$, drop $\dot{q}^2$ terms):

$$M\ddot{\vec{q}} + K\vec{q} = \vec{0}$$

Assume $\vec{q} = \vec{Q}e^{i\omega t}$: non-trivial solutions require the **characteristic equation**

$$\det(K - \omega^2 M) = 0 \quad \Rightarrow \quad \text{natural frequencies } \omega_i$$

Mode shape $\vec{Q}_i$ (up to overall scale) solves $(K - \omega_i^2 M)\vec{Q}_i = \vec{0}$; lowest ω_i = coordinates in phase, higher ω_i = out of phase.

Ignorable (cyclic) coordinates & Routhian reduction

If L has no explicit q_k -dependence ($\partial L / \partial q_k = 0$; only $\dot{q}_k$ appears), Lagrange’s equation for it collapses to a conservation law:

$$\frac{\partial L}{\partial \dot{q}_k} = B_k = \text{const} \quad (\text{conserved generalized momentum})$$

Invert for $\dot{q}_k = f(\vec{q}_{\text{rest}}, B_k)$ and use it to eliminate $\dot{q}_k$ from the remaining dynamics via the **Routhian**

$$R(\vec{q}_{\text{rest}}, \dot{\vec{q}}_{\text{rest}}; B_k) \equiv L - \dot{q}_k B_k \Big|_{\dot{q}_k \rightarrow f(\vec{q}_{\text{rest}}, B_k)}$$

$$\frac{d}{dt} \frac{\partial R}{\partial \dot{q}_j} - \frac{\partial R}{\partial q_j} = 0 \quad (j \neq k)$$

This reduces the problem by one coordinate; recover $q_k(t)$ afterward by integrating $\dot{q}_k = f(\vec{q}_{\text{rest}}(t), B_k)$.

Constraint forces via Lagrange multipliers

For M coordinates with m Pfaffian constraints $\phi_k = \sum_i a_{ki} \dot{q}_i + a_{kt} = 0$ ($k = 1, \dots, m$; a holonomic constraint left un-eliminated works the same way, with $a_{ki} = \partial \phi_k / \partial q_i$):

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_j} - \frac{\partial L}{\partial q_j} = \sum_{k=1}^m \lambda_k a_{kj}, \quad j = 1, \dots, M$$

- M equations of motion + m constraint equations $\phi_k = 0$ close the system for $\vec{q}(t)$ **and** $\lambda_k(t)$
- $\sum_k \lambda_k a_{kj}$ is exactly the constraint force conjugate to q_j — this is how the forces D'Alembert/plain Lagrange discard get recovered (e.g. rod tension, knife-edge normal force)
- Recipe:** write $L = T - V$; list a_{ki} from each ϕ_k ; write the M boxed equations plus the m constraints $\phi_k = 0$; solve algebraically for λ_k (introducing a body-frame speed like $u \equiv \dot{r} \cdot \hat{b}_1$ often decouples the algebra)

Hamiltonian Mechanics

Generalized momentum & the Hamiltonian

$$p_j \equiv \frac{\partial L}{\partial \dot{q}_j}$$

$$H(\vec{q}, \vec{p}, t) \equiv \sum_j p_j \dot{q}_j - L \quad (\text{Legendre transform of } L)$$

Invert $p_j(\vec{q}, \dot{\vec{q}}, t)$ for $\dot{q}_j(\vec{q}, \vec{p}, t)$ and substitute so H is expressed purely in $(\vec{q}, \vec{p}, t)$.

When does $H = T + V$?

Split T by degree in the $\dot{q}_j$: $T = T_2 + T_1 + T_0$ (T_2 homogeneous quadratic, T_1 linear, T_0 velocity-independent — all can depend on $\vec{q}, t$). In general

$$H = T_2 - T_0 + V \quad (\text{the Jacobi integral})$$

- Reduces to $H = T + V$ **only** when $T_1 = T_0 = 0$: $\vec{r}_i(\vec{q})$ has no explicit t -dependence (scleronomic transformation) so $T = T_2$ is homogeneous quadratic in $\dot{\vec{q}}$
- Rheonomic transformations (a prescribed $\theta(t) = \omega_0 t$, a moving support, ...) generally give $T_1, T_0 \neq 0$, so $H \neq T + V$ even when H is conserved
- H is conserved whenever $\partial L / \partial t = 0$ (no explicit t left *after* substituting any prescribed motion), regardless of whether $H = T + V$

Hamilton's canonical equations

$$\dot{q}_j = \frac{\partial H}{\partial p_j}, \quad \dot{p}_j = -\frac{\partial H}{\partial q_j}$$

$2M$ first-order ODEs replace the M second-order Euler-Lagrange equations — the natural form for numerical IVP integration. If q_k is cyclic ($\partial H / \partial q_k = 0$): $p_k = 0 \Rightarrow p_k = \text{const}$ (matches the Lagrangian/Routhian result, $p_k \equiv B_k$).

Canonical transformations & generating functions

$(\vec{q}, \vec{p}) \rightarrow (\vec{Q}, \vec{P})$ is **canonical** if Hamilton's equations hold in the new variables for some $H'(\vec{Q}, \vec{P}, t)$. A **type-1 generating function** $F(\vec{q}, \vec{Q}, t)$ produces one via

$$p_j = \frac{\partial F}{\partial q_j}, \quad P_j = -\frac{\partial F}{\partial Q_j}, \quad H' = H + \frac{\partial F}{\partial t}$$

- If F has no explicit t : $H'(\vec{Q}, \vec{P})$ is just H re-expressed in the new variables
- Goal:** pick F so H' is simple (e.g. every Q_j cyclic) — then $\dot{P}_j = 0$ and $\dot{Q}_j = \partial H' / \partial P_j = \text{const}$, trivially integrable

- Recipe:** (1) $p(q, Q) = \partial F / \partial q$; (2) $P(q, Q) = -\partial F / \partial Q$; (3) invert for $q(Q, P)$, then $p(Q, P)$; (4) substitute into $H(q, p)$ to get $H'(Q, P)$; (5) solve the (now trivial) EOM in (Q, P) ; (6) transform back for $q(t), p(t)$

1-D Dynamical Systems & Bifurcations

Fixed points & linear stability

System $\dot{x} = f(x; r)$, r a control (parameter).

$$\text{Fixed pts: } f(x^*; r) = 0 \quad \text{Stability: } \dot{\eta} = f'(x^*) \eta, \quad \eta \equiv x - x^*$$

- $f'(x^*) < 0$: stable/attracting $f'(x^*) > 0$: unstable/repelling
- $f'(x^*) = 0$: linearization inconclusive — the *marginal/degenerate* case where bifurcations live

Recipe: locate a bifurcation

- Find all fixed points $x^*(r)$: solve $f(x; r) = 0$
- Compute $f_x(x; r) = \partial f / \partial x$ and evaluate along each branch $x^*(r)$
- A bifurcation sits at (x_c, r_c) where the branch is *degenerate*:

$$f(x_c; r_c) = 0 \quad \text{and} \quad f_x(x_c; r_c) = 0$$

(solve these two equations simultaneously for x_c, r_c — this is exactly where two branches of $x^*(r)$ meet, or a branch's stability flips)

- Classify using the recipe/table below (Taylor-expand f about (x_c, r_c) if it's not already in normal form)

Fast decision path (given explicit $f(x, r)$)

- Is $x = x_0$ a root of f for *every* r (a “trivial” branch, often $x_0 = 0$)? **No** $\rightarrow$ solve $f = 0$, $f_x = 0$ together for isolated $(x_c, r_c) \rightarrow$ **saddle-node**. **Yes** $\rightarrow$ continue.
- Is $f(x, r)$ odd about x_0 , i.e. $f(x_0 - x, r) = -f(x_0 + x, r)$? **No** $\rightarrow$ **transcritical** at r_c solving $f_x(x_0, r_c) = 0$. **Yes** $\rightarrow$ **pitchfork** at that r_c ; Taylor/series-expand $f \approx (r - r_c)(x - x_0) + a(x - x_0)^3$ (expand any $\sinh, \frac{x}{1+x^2}$, etc. to cubic order) — $a < 0$ supercritical, $a > 0$ subcritical.

Classification (normal forms)

Let $\xi = x - x_c$, $\mu = r - r_c$.

- Saddle-node (fold):** $\dot{\xi} = \mu - \xi^2$ (or $\mu + \xi^2$) — *no* fixed point at $\xi = 0$ for all μ ; two fixed points ($x^* \propto \pm\sqrt{\mu}$) collide and annihilate as μ crosses 0, existing only on one side of r_c . Test: $f_r \neq 0$, $f_{xx} \neq 0$.
- Transcritical:** $\dot{\xi} = \mu\xi - \xi^2$ — $\xi = 0$ is a fixed point for *all* μ (trivial branch persists forever); a second branch $\xi = \mu$ crosses it at $\mu = 0$ and the two *exchange stability*; total # of fixed points is unchanged across r_c . Test: $f_{x\mu} \neq 0$, $f_{xx} \neq 0$.
- Supercritical pitchfork:** $\dot{\xi} = \mu\xi - \xi^3$ — symmetric ($f(-\xi) = -f(\xi)$); trivial branch $\xi = 0$ stable for $\mu < 0$, goes unstable at $\mu = 0$ and sheds *two new stable* branches $\xi = \pm\sqrt{\mu}$ for $\mu > 0$. Cubic coefficient < 0 .
- Subcritical pitchfork:** $\dot{\xi} = \mu\xi + \xi^3$ — same symmetry, but sheds *two unstable* branches for $\mu < 0$; trivial branch is unstable for $\mu > 0$ with no nearby stable state (real systems jump/hysteresis to a distant attractor via a higher-order saddle-node). Cubic coefficient > 0 .

Phase line (1-D phase portrait)

Plot $\dot{x}$ vs. x for fixed r : mark equilibria where the curve crosses $\dot{x} = 0$, arrows on the x -axis between crossings showing the sign of $\dot{x}$ (right if $f > 0$, left if $f < 0$).

- Filled dot: stable (f goes $+$ $\rightarrow$ $-$ through it, arrows point *in*); open dot (o): unstable ($- \rightarrow +$, arrows point *out*)
- Half-filled dot: semi-stable (same sign on both sides — the signature of a saddle-node exactly at $r = r_c$)
- Sweeping r and stacking phase lines side-by-side (equilibria vs. r) reconstructs the bifurcation diagram; solid branch = stable, dashed = unstable

2-D extension (nullclines, Hopf bifurcation) mirrors this, using the trace-determinant classification below to track how the fixed point's type changes with r .

2-D Linear Systems & Classification

State-space form & the Jacobian

Any $\dot{x} = f(x, \dot{x})$ becomes a 2-D first-order system with $v \equiv \dot{x}$: $\dot{x} = v$, $\dot{v} = f(x, v)$. More generally, for a pair of 1st-order ODEs $\dot{q}_1 = f_1(q_1, q_2)$, $\dot{q}_2 = f_2(q_1, q_2)$, linearizing about a fixed point $\vec{q}^*$ (where $f_1 = f_2 = 0$) with $\vec{\eta} = \vec{q} - \vec{q}^*$ gives:

$$\dot{\vec{\eta}} = A \vec{\eta}, \quad A = \begin{bmatrix} \partial f_1 / \partial q_1 & \partial f_1 / \partial q_2 \\ \partial f_2 / \partial q_1 & \partial f_2 / \partial q_2 \end{bmatrix}_{\vec{q}^*} \quad (\text{Jacobian})$$

This is the direct 2-D analogue of $\dot{\eta} = f'(x^*)\eta$ from the 1-D case above — same idea, now with a matrix.

Eigenvalues, trace & determinant

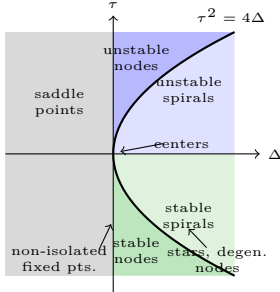
Try $\vec{\eta} = \vec{v} e^{\lambda t}$: non-trivial solutions require $\det(A - \lambda I) = 0$, i.e.

$$\lambda^2 - \tau\lambda + \Delta = 0 \Rightarrow \lambda_{1,2} = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2}$$

$$\tau \equiv \text{tr } A = \lambda_1 + \lambda_2, \quad \Delta \equiv \det A = \lambda_1 \lambda_2$$

General solution (distinct λ_i): $\vec{\eta}(t) = c_1 \vec{v}_1 e^{\lambda_1 t} + c_2 \vec{v}_2 e^{\lambda_2 t}$. Stability depends only on $\text{Re}(\lambda_{1,2})$ — i.e. only on (Δ, τ) — so the fixed point can be classified *without* solving for eigenvectors.

Classification: the (Δ, τ) plane



- $\Delta < 0$ — **saddle**: real, opposite-sign eigenvalues ($\lambda_1 \lambda_2 = \Delta < 0$) $\Rightarrow$ *always unstable* (one stable + one unstable eigendirection), regardless of τ
- $\Delta > 0, \tau^2 > 4\Delta$ — **node**: real eigenvalues, same sign (sign of τ , since $\Delta > 0$). $\tau > 0$: unstable node; $\tau < 0$: stable node
- $\Delta > 0, \tau^2 < 4\Delta$ — **spiral**: complex-conjugate pair $\lambda = \frac{\tau}{2} \pm i\omega$; real part $\tau/2$ sets the growth/decay of the oscillatory envelope. $\tau > 0$: unstable spiral; $\tau < 0$: stable spiral
- $\tau = 0, \Delta > 0$ — **center**: $\lambda = \pm i\sqrt{\Delta}$ purely imaginary $\Rightarrow$ neutrally stable closed orbits *for the linear system only* — nonlinear terms generically destroy a center (turn it into a weak spiral)
- $\tau^2 = 4\Delta$ (**on the parabola**), $\Delta > 0$ — **repeated real $\lambda = \tau/2$** : *star node* if $A \propto I$ (two independent eigenvectors, every direction is an eigendirection); *degenerate node* otherwise (only one eigenvector; trajectories become tangent to it)
- $\Delta = 0$ — **non-isolated fixed points**: at least one $\lambda = 0 \Rightarrow$ a whole *line* of fixed points, not just the origin. $\tau < 0$ (other eigenvalue negative): line attracts; $\tau > 0$: line repels

Recipe: (1) find fixed point(s) $\vec{q}^*$ solving $f_1 = f_2 = 0$; (2) build the Jacobian A there; (3) compute $\tau = \text{tr } A$, $\Delta = \det A$; (4) locate (Δ, τ) in the plane above to classify — no need to solve for λ just to determine type/stability; (5) only solve for $\lambda_{1,2}$ (and eigenvectors, if sketching the actual phase portrait/eigendirections) after classifying.

Eigenvectors & the Hartman–Grobman theorem

Saddle arrow directions: solve $(A - \lambda_i I)\vec{v}_i = \vec{0}$ for each λ_i — the negative- λ eigenvector is the incoming (stable) direction, the positive- λ eigenvector the outgoing (unstable) direction; these are the only directions trajectories can approach/leave along and they organize the global separatrices.

Hartman–Grobman: if every λ_i has $\text{Re}(\lambda_i) \neq 0$ (a *hyperbolic* fixed point), the nonlinear flow near $\vec{q}^*$ is topologically equivalent to the linearized flow $\dot{\vec{\eta}} = A\vec{\eta}$ — linearized stability *is* nonlinear stability.

The guarantee **fails** exactly when some $\text{Re}(\lambda_i) = 0$: a **center** ($\tau = 0, \Delta > 0$) is only guaranteed in the full nonlinear system if it has an exact conserved quantity (e.g. energy) — otherwise nonlinear terms can turn it into a weak spiral; $\Delta = 0$ (**non-hyperbolic**, a zero eigen-

value) also voids the theorem and typically signals a line/continuum of fixed points.

Full pipeline: classify & sketch a nonlinear phase portrait

1. **State space**: for $\dot{x} = f(x, \dot{x})$ set $v \equiv \dot{x}$ to get $\dot{x} = v$, $\dot{v} = f(x, v)$; in general $\dot{q}_1 = f_1(q_1, q_2)$, $\dot{q}_2 = f_2(q_1, q_2)$
2. **Fixed points**: solve $f_1 = f_2 = 0$ simultaneously
3. **Linearize once, generally**: build $A(q_1, q_2)$ symbolically (partials above)
4. **Evaluate A at each fixed point separately** — linearization is only locally valid, so every fixed point gets its own numeric Jacobian and its own classification
5. **Classify** each via (Δ, τ) per the table above
6. **Eigenvectors** for saddle arrow directions — solve $(A - \lambda I)\vec{v} = \vec{0}$ per the box above
7. **Sketch**: plot each fixed point with its local behavior (node/spiral/saddle/center); nullclines $\dot{x} = 0$, $\dot{v} = 0$ partition the plane into regions of constant flow sign (trajectories never cross them except at fixed points); connect saddle separatrices to assemble the global portrait
8. **If a parameter r is present**, repeat steps 2–7 across r — a bifurcation occurs exactly where an eigenvalue crosses zero ($\Delta = 0$) or the imaginary axis ($\tau = 0$ with $\Delta > 0$, a **Hopf bifurcation**), the direct 2-D analogue of the 1-D bifurcation recipe above

Nondimensionalizing an Equation of Motion

Recipe

1. Write the dimensional EOM and list every dimensional quantity appearing in it (state variable(s), t , and all parameters: $m, k, b, g, \ell, \dots$)
2. Choose a characteristic **length** L and **time** T built from the problem's own parameters (e.g. L = a natural length/amplitude in the geometry, $T = \sqrt{m/k}$ or $1/\omega_n$) — these are what get scaled out
3. Define dimensionless variables $\xi = x/L$, $\tau = t/T$ (one pair per independent physical dimension in play; add a scaled force/energy if one appears)
4. Convert derivatives by chain rule: $\dot{x} = \frac{L}{T}\xi'$, $\ddot{x} = \frac{L}{T^2}\xi''$ (primes = $d/d\tau$)
5. Substitute into the EOM; divide the whole equation by the coefficient of the *highest* derivative so it has unit leading coefficient
6. Collect every remaining combination of parameters into as few dimensionless groups as possible (Buckingham π : #groups = #parameters – #independent dimensions); name each one ($\zeta, \epsilon, \dots$) — these are the only “knobs” left in the problem
7. Sanity check: every leftover coefficient multiplying a ξ, ξ', ξ'' term must be a pure number (no residual kg/m/s)

Worked template (spring–mass–damper type)

Dimensional: $m\ddot{x} + b\dot{x} + kx = F(t)$ (e.g. a gravity component or drive). With $T = \sqrt{m/k}$, L = a natural length scale, $\tau = t/T$, $\xi = x/L$:

$$\xi'' + 2\zeta\xi' + \xi = \tilde{F}(\tau), \quad 2\zeta \equiv \frac{b}{\sqrt{mk}}, \quad \tilde{F} \equiv \frac{FT^2}{mL}$$

ζ (damping ratio) is the one leftover dimensionless group here; a tilted/geometric system typically adds one more (an angle α — already dimensionless — or a length ratio a/L), matching “3 dimensionless parameters” style results.

Overdamped/singular limit $\Rightarrow$ drops the order by one

If a derivative's dimensionless coefficient is formally small (heavy damping $\Rightarrow$ small effective inertia, or a genuinely small mass), write it as $\epsilon\xi'' + \xi' + \dots = 0$ with $\epsilon \ll 1$ and set $\epsilon \rightarrow 0$ (quasi-static/regular-perturbation assumption):

$$\epsilon\xi'' + \xi' = g(\xi; \vec{p}) \xrightarrow{\epsilon \rightarrow 0} \xi' = g(\xi; \vec{p})$$

This collapses a 2nd-order (2-D phase-plane) system to a **1-D, 1st-order** system — the phase-line/bifurcation analysis above applies directly, with the remaining dimensionless groups $\vec{p}$ playing the role of τ .